Mentl: proof-carrying control software for the physical world

Hard Tech Week 2026 · Nucleus × Startup State open call · [[name]] · sonofagoose@pm.me · github.com/ampactor-labs/mentl · github.com/ampactor-labs/sonido

THE PROBLEM

Software that steers physical things (inverters, flywheels, flight controls) is tested, reviewed and hoped on: no mainstream language lets a team state "this loop never reaches the network, never allocates, never commands past this bound" and have the compiler prove it. Coding agents now write that software, and the grid is where the gap bites first: every new inverter is a software product under UL 1741 SB, and Utah's data centers are the load.

WHAT EXISTS TODAY

Mentl is a programming language and self-hosting compiler with an algebraic effect system, compiled to WebAssembly. A negated effect such as `!Network` is a claim proven through every call or the program does not build; every fact carries its reason; a learning rule is derived from the chain it learns on. The pinned compiler reproduces itself byte for byte. Seven months and 2,600 commits of work, open source (Apache and MIT). Beside it, sonido: a Rust DSP framework whose kernels run unchanged in plugins, a browser editor and a guitar pedal.

BUILT THIS WEEK (SINCE MONDAY 4:00 PM)

- HUM: the grid's frequency heard live from a room's wiring through a guitar pickup, read to the millihertz three independent ways; a Torus Station modeled from Torus's published figures answering in two bands, seen planning around tomorrow's rush hour with the clock set; a real 2019 grid disturbance replayed from open data.
- The controller written in Mentl: proven unable to make any host call while deciding (no network, no file, no clock), to allocate mid-decision, or to command either machine past its rating; every decision explained in words; a twin with one price lookup added, refused with the path named.
- The learner, whose gradient the compiler derives; the page; the ear in the pedal; an agent's proposal refused by the proof gate.

WHO NEEDS IT, IN UTAH

- **Torus** (South Salt Lake): the Station hears every cycle; the program beneath the dispatcher has never been provable. The Torus judge is seeing this demo at the same hackathon. First conversation: Torus engineering, this month.
- **Hill AFB, 309th Software Engineering Group**: one of the Air Force's largest software organizations, maintaining flight software for the F-16, the F-35 and the missile fleet; assurance evidence is a deliverable.

MILESTONES

Day 30	The proven controller on inverter-class hardware on the bench. The company formed [[date / status]]; SAM.gov, SBIR.gov and DSIP registrations filed on day one, since they take weeks and cost nothing.
Day 60	One Utah site hosting an ear on a Station-class inverter, the dispatch explained to its operator: Torus's if that door opens, another's if not. The SBIR topic chosen with Nucleus Grow (AFWERX open topic): an assurance tool for a flight-software module, the 309th's need in their words.
Day 90	A pilot letter from the day-60 site; ninety days of its decisions on the page; the Phase I draft ready for the next open-topic window.

THE $10,000

$7,000	inverter-class development hardware: a three-phase inverter platform and its DC supply, the controller on its supervisor
$3,000	the proof gate hosted for ninety days, $1,000 a month, for the pilot site and the first team running coding agents through it

PEOPLE AND RESOURCES I NEED

- One Utah storage or inverter operator who will host an ear: Torus, if that door opens.
- The 309th, through 47G, and Nucleus Grow for the SBIR lane and the Phase I draft.
- A co-founder with power-electronics depth, when you meet one. Connecting people is what this program does; this is my request.

Business: paid pilots with asset operators first; the proof gate for teams running coding agents second; the SBIR lane is non-dilutive runway for the assurance tool. The demo: `github.com/ampactor-labs/hum`; the proofs: `mentl check`. [[one line: who you are, in your words]]